INN Hotels Booking Cancellation Prediction

Supervised Learning

08/03/2025

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Executive Summary

*** Actionable insights and recommendations***

- Bookings with short lead times are important since they don't have many cancellations. Optimize the process for last minute bookings.
- Repeated customers can reduce the possibility of cancellations. Customers should be given incentives to visit again.
- Customers can be given incentives in terms of special requests fulfillment as well as attractive meal plans. When they select these, there is a less chance of them canceling.
- Adding a cancellation fee may prevent customers from canceling for other cheaper hotel rooms. This fee needs to be priced correctly. More analysis may be needed on how much should it be.
- Offer promotions for all market segments - online, offline and aviation to reduce cancellations.

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Business Problem Overview and Solution Approach

Problem overview:

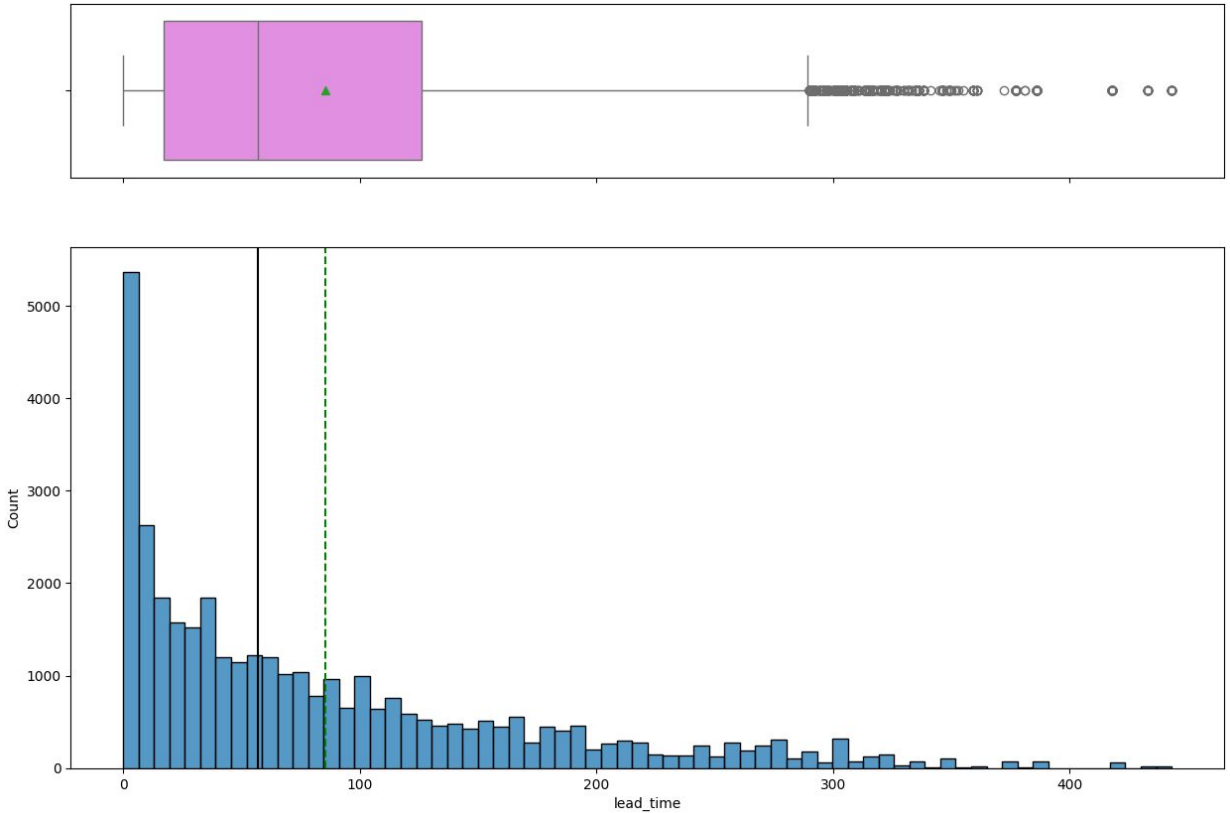
- ★ A significant number of hotel bookings are called off due to cancellations or no-shows.
- ★ High cancellation rates cause operational inefficiencies and lower revenue.
- ★ A machine learning based solution is needed to predict which booking will result in cancellations. This can boost revenue and increase efficiency.

Solution/methodology:

- ★ Provide a machine learning solution to the problem that can predict the booking that may result into cancellations. This may include following steps:
  - ○ EDA - Univariate and bivariate analysis of the data to understand the booking and cancellation patterns.
  - ○ Predictive modeling- Logistic regression and decision tree modeling to provide a framework for predicting cancellations of bookings.
  - ○ Comparison of the models and choose the best one.
  - ○ Develop business strategies based on the models and EDA to improve the business practices, improve efficiencies and increase revenue as well as customer satisfaction.

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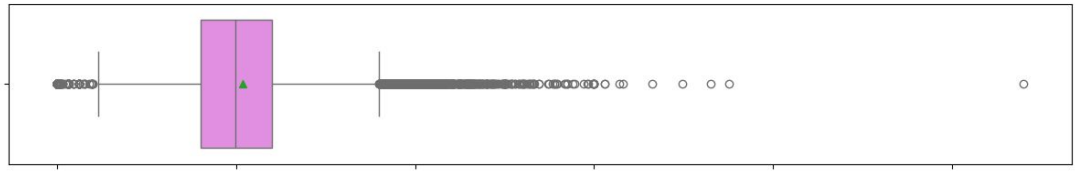
EDA Results - Lead time

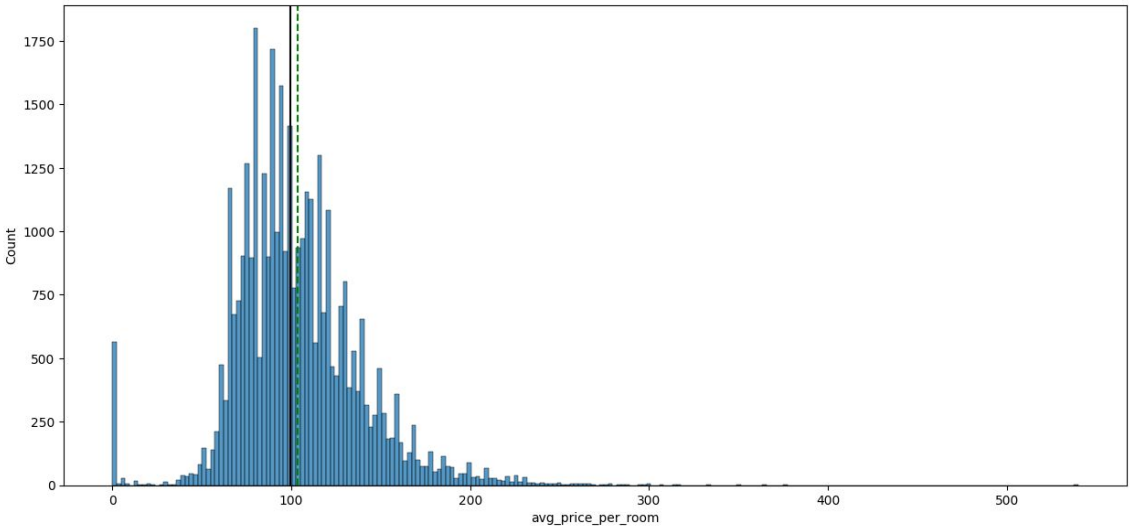


Most customers book their stay closer to the actual date of arrival.

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EDA Results - Average price per room

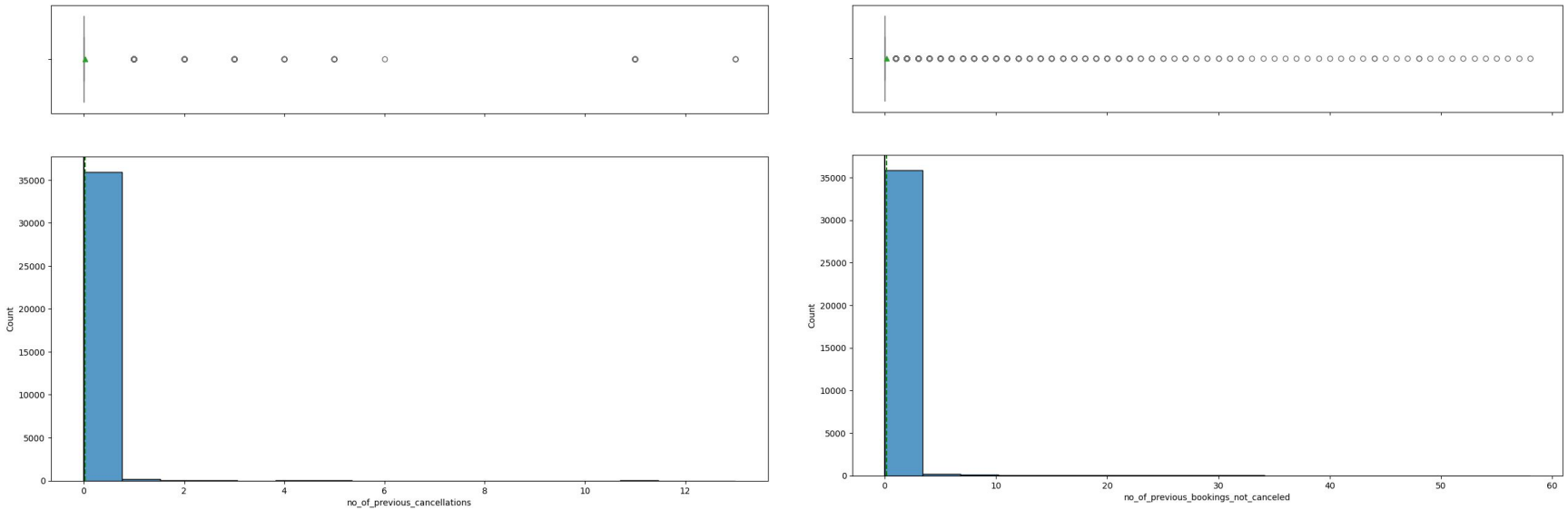




Average room price is concentrated between 50 to 150. Higher room prices are less common.

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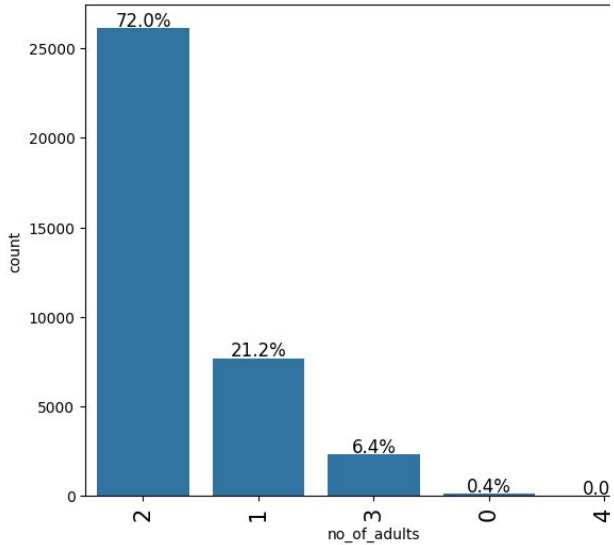
EDA Results - Previous Cancellation rates

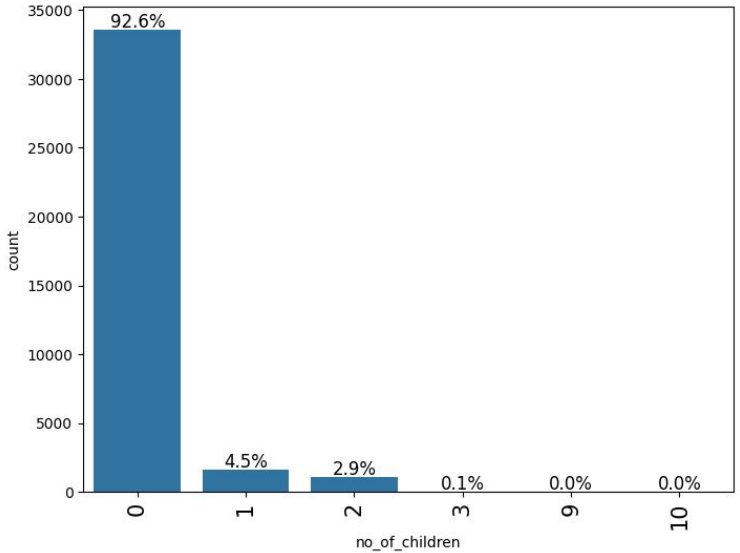


Majority customers don't have previous cancellations. But very few customers have high previous cancellation rate.

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EDA Results - Adults Vs Children

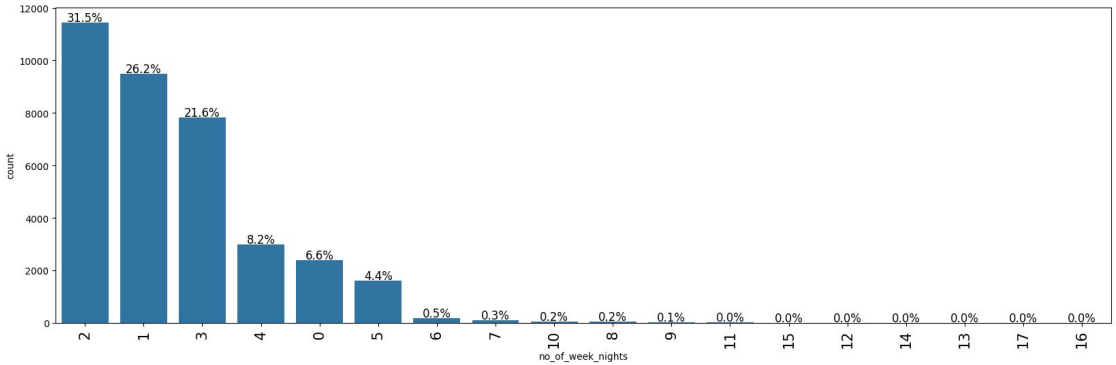


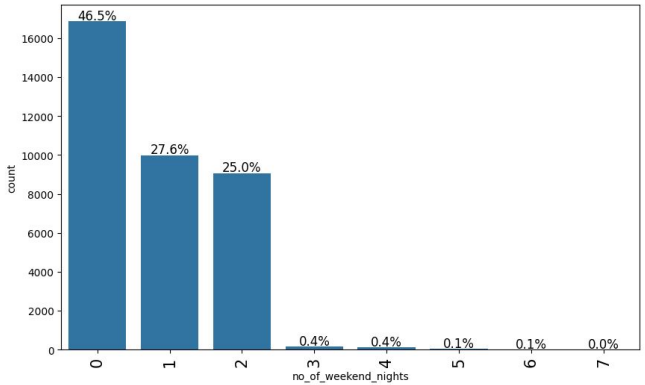


Two adults and no children is most common in hotel bookings.

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EDA Results - Weekend Vs Week nights

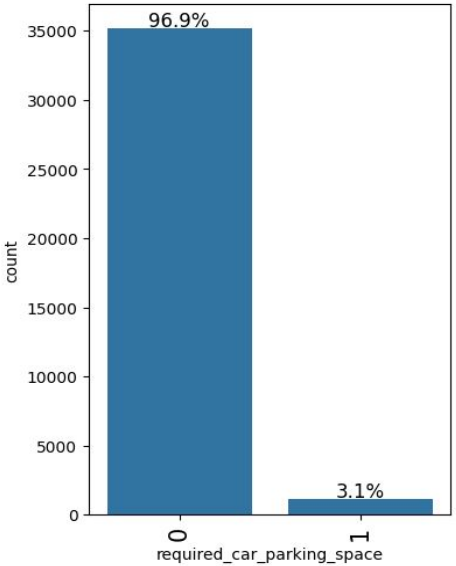


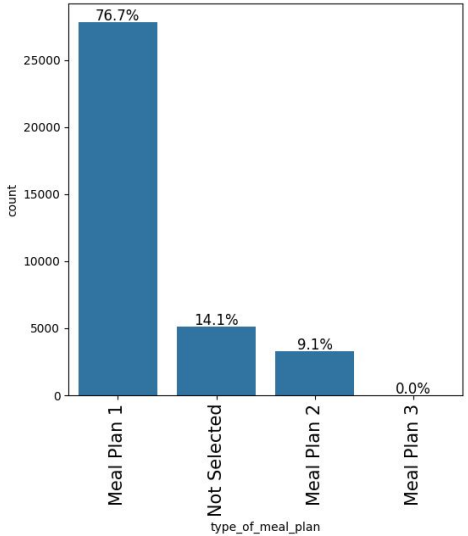


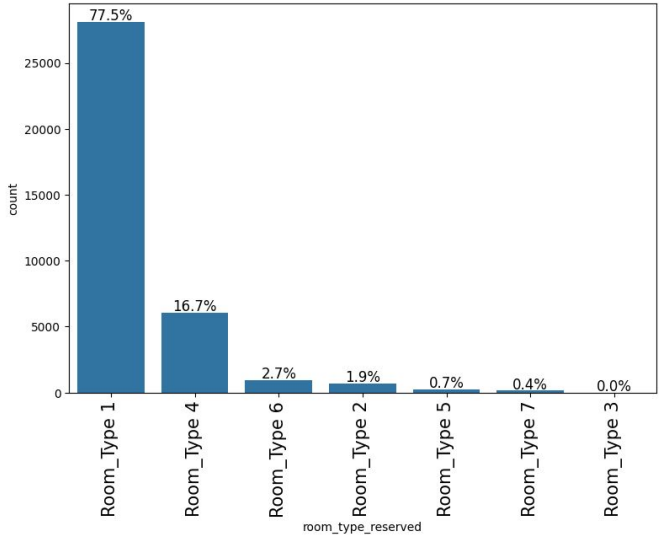
On weekdays, customers prefer to stay between 1 to 3 nights. Weekdays are preferred to weekends. Weekend nights stays are 1 or 2 nights.

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EDA Results - preferences



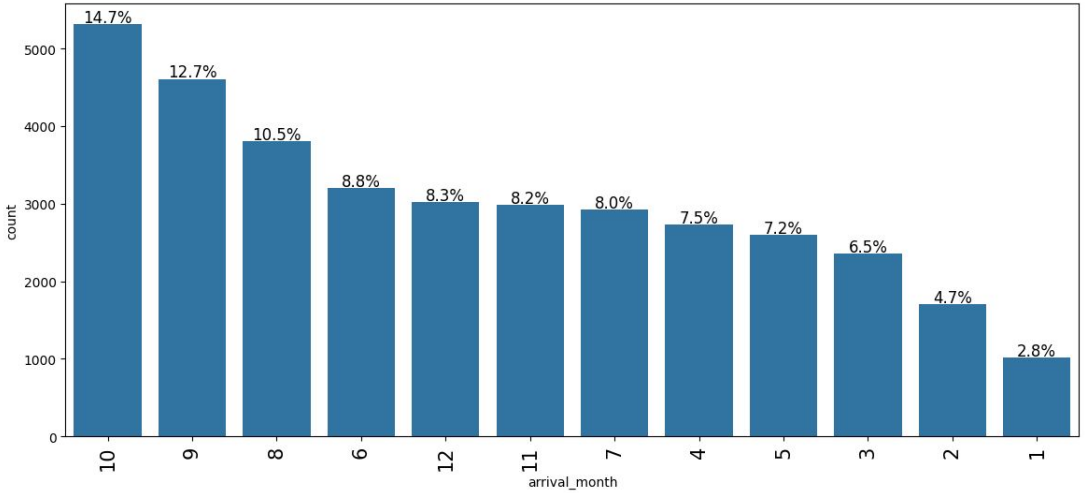


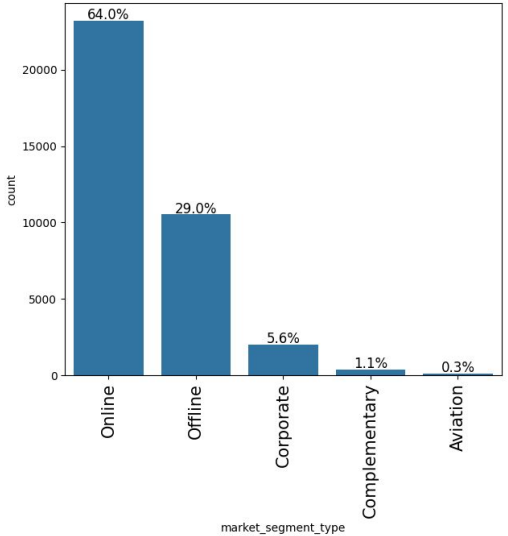


Most customers don't need car parking space, also, most customers choose meal plan 1 and room type 1.

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EDA Results - bookings



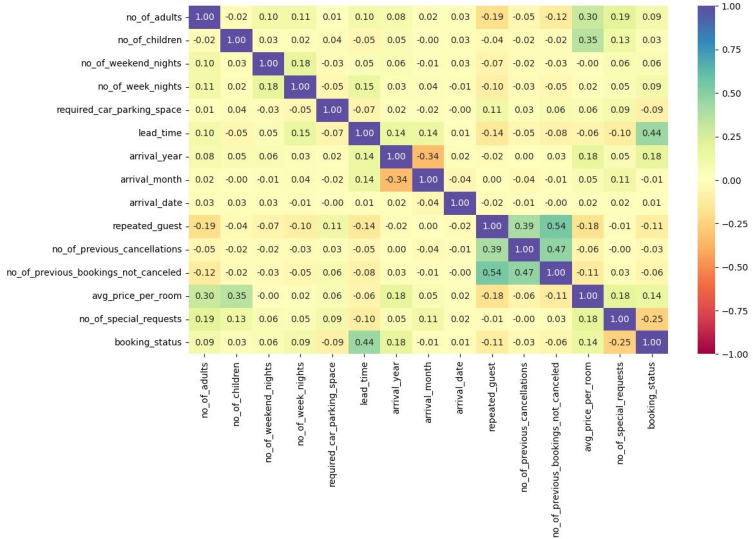


September-October is high season for hotel bookings. Most bookings are online.

Most customers do not cancel the bookings, but 32% bookings do get cancelled. Half of the customers have no special requests, but other half have one or two special requests.

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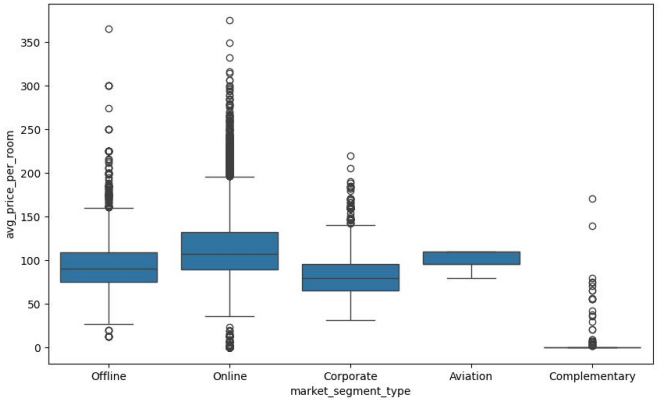
EDA Results - Bivariate analysis

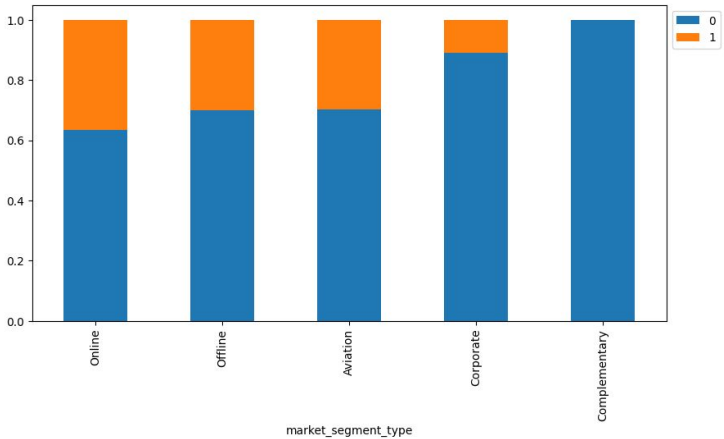


- Longer lead times are associated with higher cancellation rates.
- Negative correlation between the number of previous cancellations and booking status
- Guests with more special requests are less likely to cancel.
- Repeated guests are less likely to cancel.
- If number of adults is more, the lead time is more.

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EDA Results - Prices vs Market segments

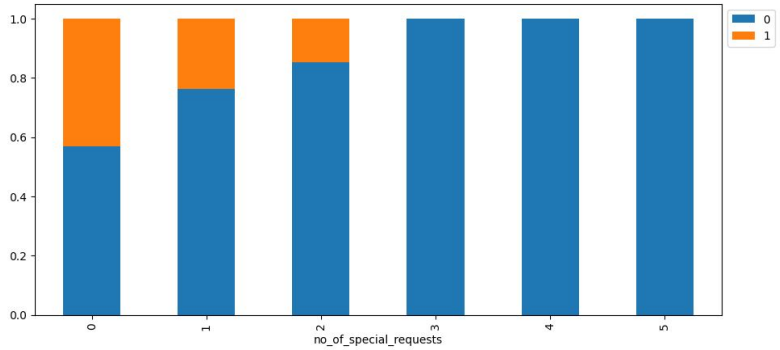


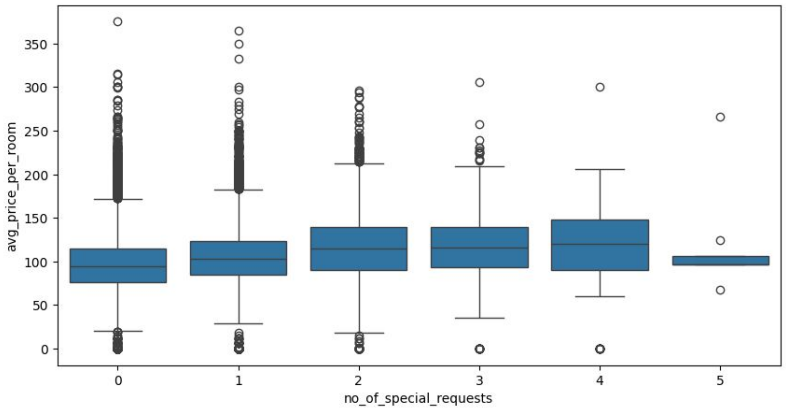


- Offline and online market segments show wide variability in prices. They also have more cancellations.
- Corporate is more consistent in price and very less cancellations.
- Aviation has no outliers. Cancellations are less than online but same as offline.
- Complementary rooms price is low. There are no cancellations.

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EDA Results - Special requests

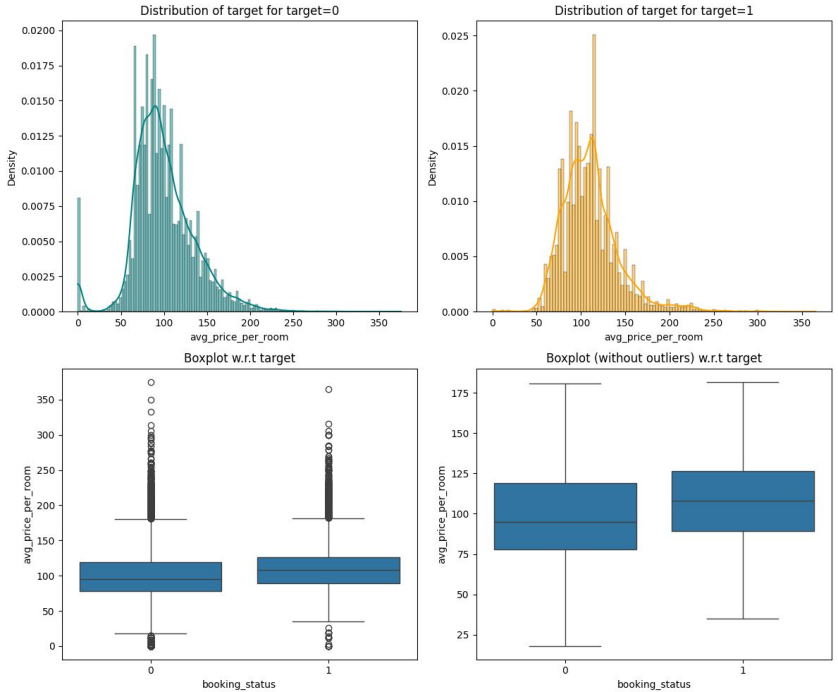




- Likely cancellations when no special requests are made.
- More number of special requests reduces chance of cancellations.
- Median price of rooms does not change a lot with more number of special requests, it only increases slightly.

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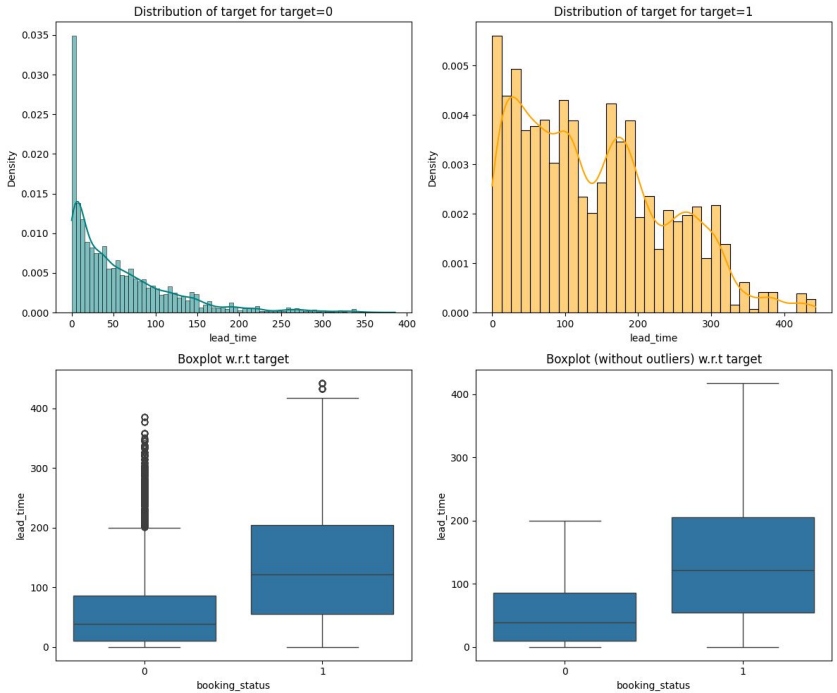
EDA Results - Booking status and average price per room



- Distribution of Prices for Canceled Bookings (target=1)- slightly skewed
- Distribution of Prices for non-canceled Bookings (target=0)- have more higher prices.
- Price does not seem to be a predictor of cancellations.

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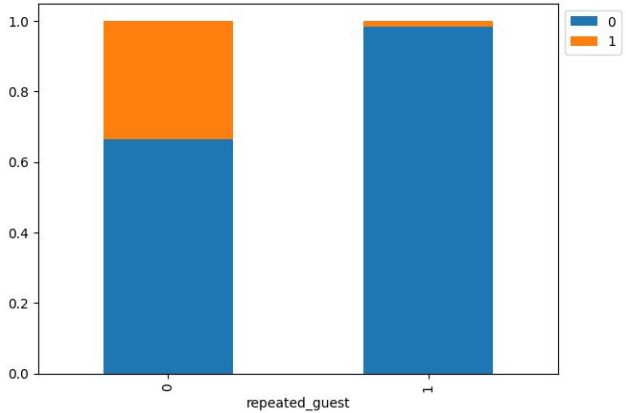
EDA Results - Lead time and booking status

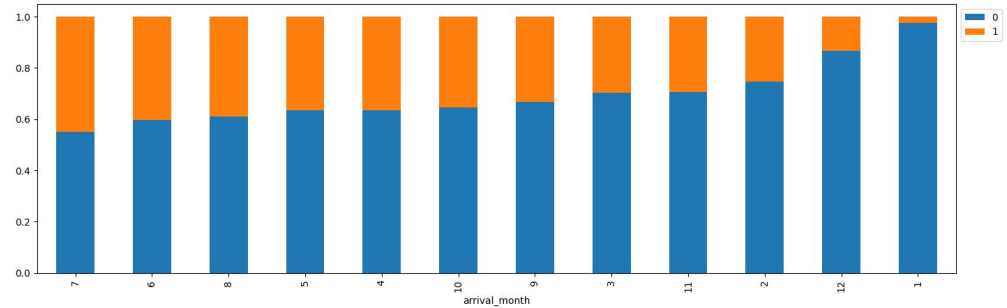


- Longer lead times correlate with higher cancellation rates
- Shorter lead time correlates with low cancellations.
-

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EDA Results - Repeated guests and seasonality in cancellations





- Repeated guests shown by the 1 column on the x axis have very less cancellations compared to new guests.
- The high season for bookings is September-October.
- June to August (summer months) have high cancellations.

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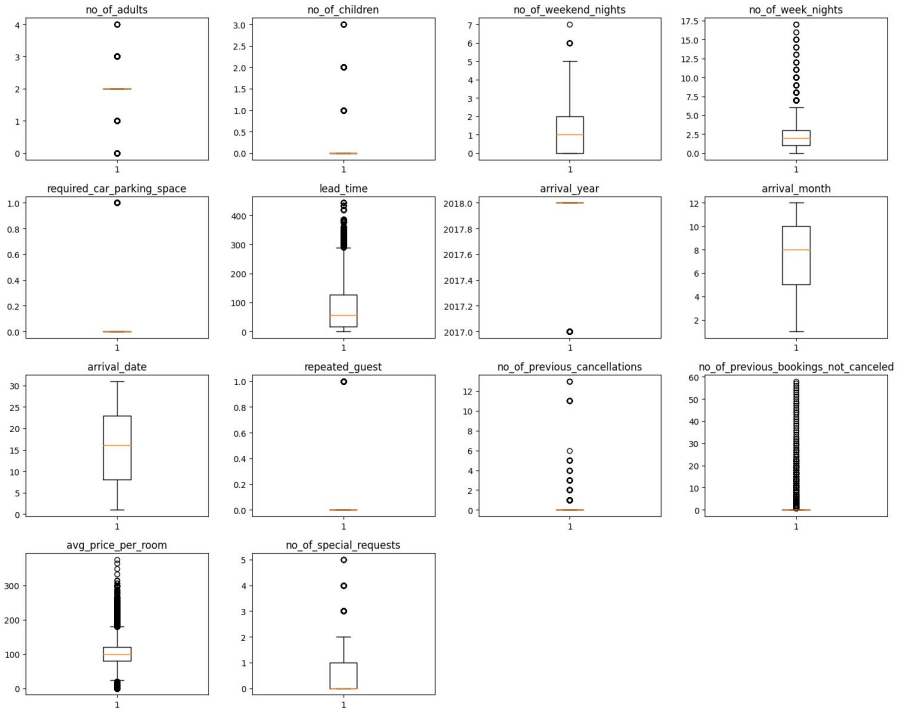
Data Preprocessing

- Duplicate value check
- Missing value treatment
- Outlier check (treatment if needed)
- Feature engineering
- Data preparation for modeling

***Note**: You can use more than one slide if needed*

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Outlier check



These box plots show the outliers of various fields with respect to the booking status. The important ones to look at are the correlations between these and booking status:

- No of adults/children in the booking
- Stay duration
- Car parking space required
- Lead time
- Arrival date
- Room price
- Special requests

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Model evaluation criterion given

Model can make wrong predictions as:

1. Predicting a customer will not cancel their booking but in reality, the customer will cancel their booking.
2. Predicting a customer will cancel their booking but in reality, the customer will not cancel their booking.

Which case is more important?

- Both the cases are important as:
- If we predict that a booking will not be canceled and the booking gets canceled then the hotel will lose resources and will have to bear additional costs of distribution channels.
- If we predict that a booking will get canceled and the booking doesn't get canceled the hotel might not be able to provide satisfactory services to the customer by assuming that this booking will be canceled. This might damage the brand equity.

How to reduce the losses?

- Hotel would want **F1 Score to be maximized,** greater the F1 score higher are the chances of minimizing False Negatives and False Positives.

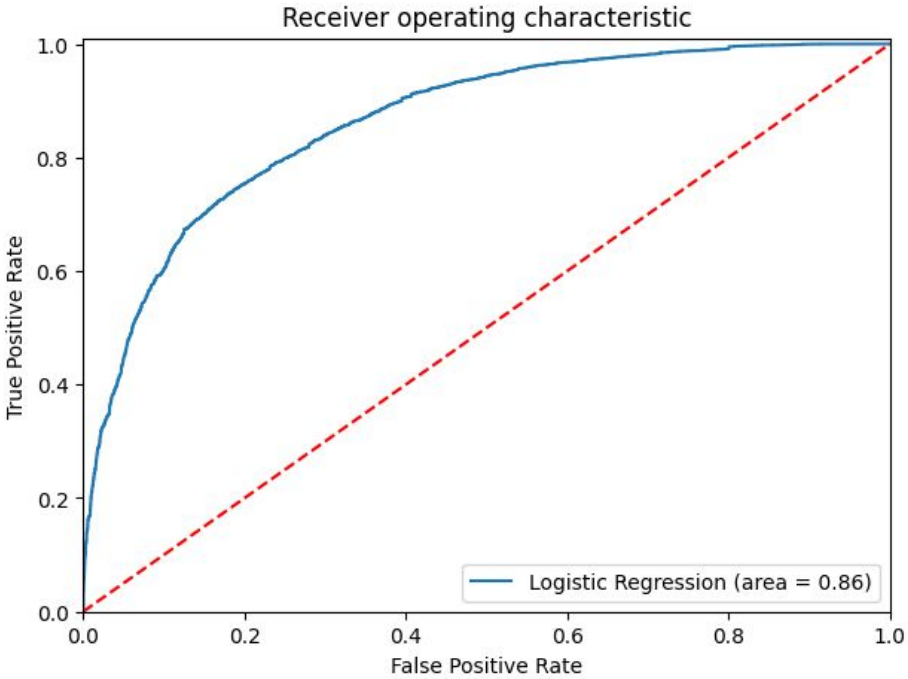
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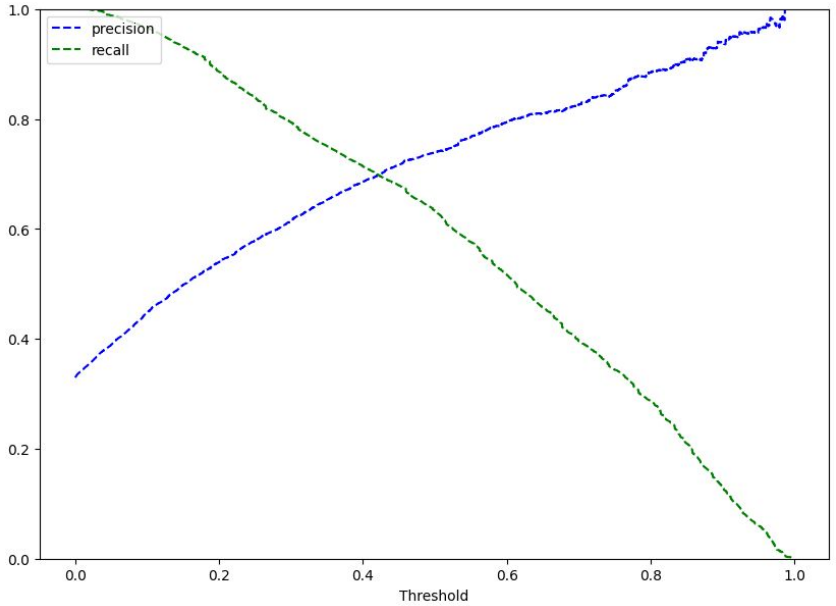
Logistic regression model

- We want to predict which bookings will be canceled.
- Before we proceed to build a model, we'll have to encode categorical features. Created dummy variables to represent the categorical variables.
- Data was split into train and test (70%,30%) to be able to evaluate the model that we build on the train data.
- Initial model was built on the training data.
- Dropped the predictor variables one by one- having a p-value greater than 0.05 as they do not significantly impact the target variable.
- The coefficients of the logistic regression model were converted to odds.
- Recall score was improved further, by changing the model threshold using AUC-ROC Curve.
- Precision-Recall curve was used to determine a better threshold.
- Optimal threshold was set to 0.42
- Performance was compared on training and testing data with three threshold values: default, 0.37 and 0.42
- 0.37 threshold model seems to be the best choice of the three- it is the most consistent across test and training and has best F1.

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AUC-ROC curve and threshold





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Logistic Regression Model Performance thresholds comparison

Training performance comparison:

	Logistic Regression-default Threshold	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.80557	0.79289	0.80139
Recall	0.63303	0.73419	0.69891
Precision	0.73918	0.66914	0.69833
F1	0.68200	0.70015	0.69862

Test performance comparison:

	Logistic Regression-default Threshold	Logistic Regression-0.37 Threshold	Logistic Regression-0.42 Threshold
Accuracy	0.80447	0.79629	0.80327
Recall	0.63118	0.73822	0.70216
Precision	0.72837	0.66752	0.69369
F1	0.67630	0.70109	0.69790

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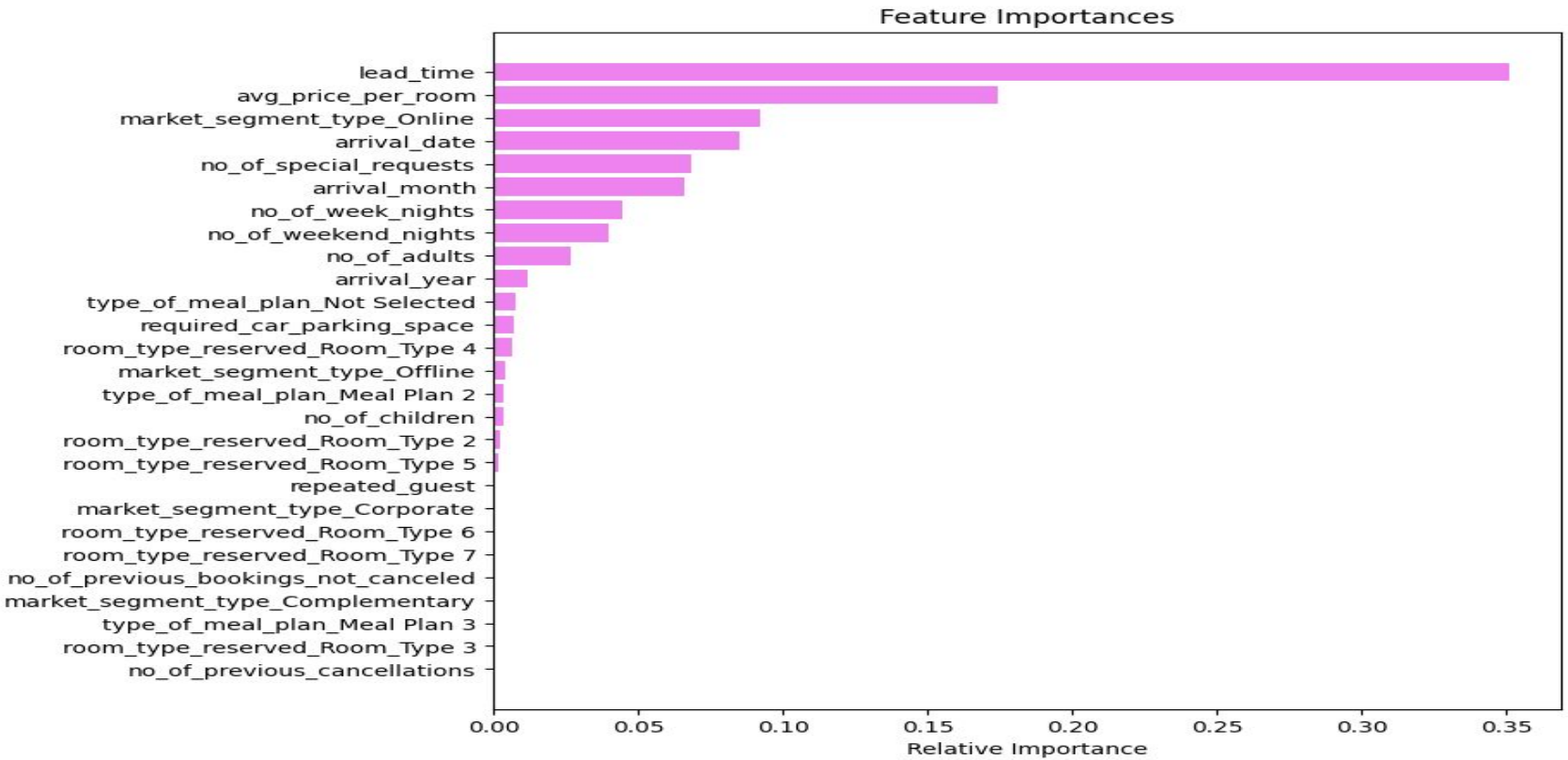
Decision Tree Model

- We want to predict which bookings will be canceled.
- Encoded the categorical features with dummy variables.
- Split the data into train and test (70%, 30%) to be able to evaluate the model that we build on the train data.
- Fit the decision tree on training data.
- Checked the performance on the training data and test data.
- Important features were checked.
- Pre-pruning the tree - hyperparameter tuning was performed with these parameters
  - `"max_depth": np.arange(2, 7, 2),`
  - `"max_leaf_nodes": [50, 75, 150, 250],`
  - `"min_samples_split": [10, 30, 50, 70],`
- Checked the model performance again on training and testing data - after pre-pruning.
- Decision tree visualization was done and important features were checked.
- Cost complexity pruning was done
- Checked the performance on the training data and test data.
- Decision tree model performance was compared- initial, after pre-pruning and after post-pruning.
- Post pruning seems to yield consistent and good results across test and training.

Link to Appendix slide on data background check

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Feature importances



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Decision Tree Model Performance comparison

Training performance comparison:

Decision Tree sklearn	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)	
Accuracy	0.99421	0.99421	0.89954
Recall	0.98661	0.98661	0.90303
Precision	0.99578	0.99578	0.81274
F1	0.99117	0.99117	0.85551

Test performance comparison:

Decision Tree sklearn	Decision Tree (Pre-Pruning)	Decision Tree (Post-Pruning)	
Accuracy	0.87118	0.87118	0.86879
Recall	0.81175	0.81175	0.85576
Precision	0.79461	0.79461	0.76614
F1	0.80309	0.80309	0.80848

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Model Performance Summary

- Best logistic regression model: threshold - 0.37 .This is chosen as final because F1 is highest and results are consistent in training and testing data.
- Best decision tree model: Post pruning. This is chosen as final because results are consistent across testing and training and it is providing a big improvement in F1.
- **Final chosen model is: Post Pruning Decision Tree Model. This is a winner in all metrics.**

	Logistic Regression-0.37 Threshold training	Logistic Regression-0.37 Threshold testing	Decision tree post pruning training	Decision tree post pruning testing
Accuracy	0.79289	0.79629	0.89954	0.86879
Recall	0.73419	0.73822	0.90303	0.85576
Precision	0.66914	0.66752	0.81274	0.76614
F1	0.70015	0.70109	0.85551	0.80848

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